

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2BI2__ / 2IPBI2__

REGISTRATION
NUMBER

BIOLOGY

9648/02

Paper 2

26 August 2015

2 hours

Additional Materials: Answer Paper and Cover Page

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you submit.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

The number of marks is given in brackets [] at the end of each question or part question.

Section A to C

Answer all questions.

Write your answers in this booklet.

Section D

Write your answers on the separate Answer Paper.

Begin each section of the question on a new piece of paper.

For Examiner's Use	
Section A - C	[Total: 80]
1	10
2	10
3	10
4	10
5	10
6	10
7	10
8	10
Section D	[Total: 20]
9 / 10	20
Total	/ 100

This document consists of **23** printed pages, including this cover page and one blank page.

Section A

Answer **all** the questions in this section.

- 1 In beer-making, enzymes present in malting barley hydrolyse starch into sugar, ready for fermentation.

- (a) Compare the structure of starch and cellulose, and their roles in plants.

[3]

Starch	Cellulose
<u>Storage role in plants</u> where excess glucose made during photosynthesis is converted to starch (which does not affect solute potential of cells)	<u>Structural role</u> in plants as cellulose cell wall provide support to plants
<u>α-glucose</u> monomers	<u>β-glucose</u> monomers
<u>Alternate units not oriented 180° to each other</u>	<u>Alternate unites are oriented 180° to each other</u>
<u>Absence of hydrogen cross links</u> in starch	<u>Parallel chains are linked by hydrogen cross-links</u>
<u>No microfibril and macrofibril</u> formation in starch	<u>Microfibrils (and macrofibril)</u> formation through association of many cellulose molecules
<u>Unbranched through α (1 $\rightarrow$ 4), Branched through α (1 $\rightarrow$ 6)</u>	<u>Unbranched through β (1 $\rightarrow$ 4)</u>

- (b) The graph below shows the production of sugar at three different temperatures over a period of 60 minutes. All other conditions were controlled.

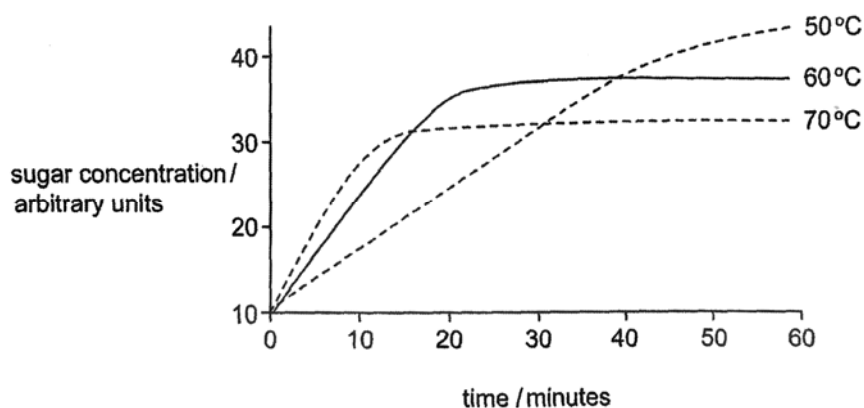


Fig. 1.1

- (i) State the enzymes used in the reaction above.

[1]

- Amylase

- (ii) Explain how the enzyme stated in (i) plays its role in hydrolysis of starch to glucose.

[3]

- Amylase contains an **active site**, containing contact amino acids which are

involved in complementary binding to the substrate

[Max. 1 mark]

- **Formation of enzyme-substrate complex (E-S complex)**
- **Lowers activation energy** / energy required for substrate to react **by**
 - o Serve as template to position substrate molecules in correct orientation for catalysis
 - o Induces stress in bonds of substrates
 - o Increases substrate reactivity

[Max. 1 mark]

- **Enzyme specificity:**
 - o **Lock and key hypothesis** – shape of substrate is complementary to the shape of the active site of enzyme
 - o **Induced Fit hypothesis** – initial binding of substrate to enzyme causes a conformation change in the shape of the enzyme which leads to more effective binding

[Max. 1 mark]

(iii) With reference to Fig. 1.1, explain the effect of increasing temperature on enzyme activity for the first 10 minutes of the reaction. [3]

- **Reaction at 70°C had the highest concentration of glucose formed per unit time** → 60°C → 50°C
- Ref: Rate of enzyme reaction increases as temperature increases
- **Increase in kinetic energy** of enzymes and substrate molecules;
- **Increase in frequency of effective collisions** between enzyme and substrate;
- Ref: Increase in molecules with sufficient energy to overcome activation energy
- Ref: Increase in concentration of enzyme-substrate complexes produced per unit time;
- Ref: Increase in concentration of products formed per unit time;

[Any 6 points → 3 marks]

[Total: 10 m]

- 2 (a) DNA is replicated semi-conservatively.
State what is meant by semi-conservative replication.

[3]

- DNA molecule separates into single strands
- By breaking hydrogen bonds between deoxyribonucleotides
- Both strands act as templates for DNA synthesis
- Each new DNA molecule contains one new and one original strand of DNA

- (b) In bacteria, it is common for two or more structural genes to be arranged together in an operon. Fig. 2.1 shows the structure of the *lac* operon and its control elements. Four different configuration of proteins binding to the control elements have been drawn for you. Identify the four conditions that will result in configuration **A**, **B**, **C**, and **D**. (Please only use the keywords: high, low, unavailable).

[4]

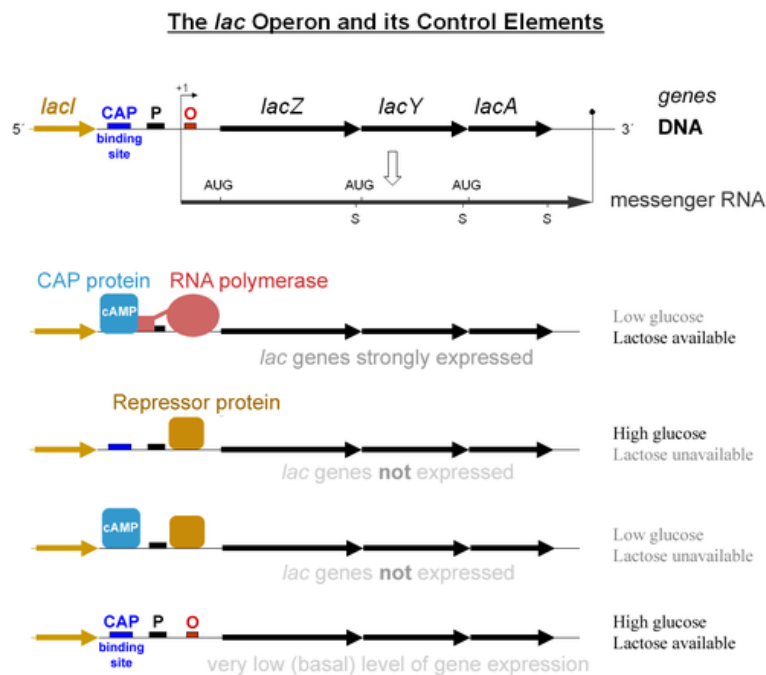


Fig. 2.1

- (c) (ii) A mutation that inactivates the *lac* repressor and prevents it from binding to the *lac* operator has been found. Explain the effect of this mutation on *lac* operon expression.

[2]

- A mutation that prevented the *lac* repressor from binding to the operator would make the *lac* operon constitutive only in the absence of glucose.
- However this mutation would not be entirely constitutive because transcription would be inhibited in the presence of glucose.

- (iii) State the disadvantage to the bacterium for having this mutation.

[1]

- Bacterial cell would waste a lot of energy to transcribe the gene and translating the mRNA when lactose was not present.

[Total: 10 m]

- 3 (a) Control of gene expression in a eukaryote can take place at different levels. Briefly explain the control of gene expression at the three different level stated.

(i) Chromosomal level

[2]

- *Eukaryotic DNA is associated with histones / form chromatin*
- *DNA methylation / demethylation and histone acetylation/deacetylation can occur*
- *Conversion of euchromatin to heterochromatin*

(ii) Transcriptional level

[2]

- *Complex interactions between protein factors and DNA sequences upstream of a gene to stimulate and initiate transcription*
- *e.g. promoters, enhancers and silencers*

(iii) Post-translational level

[2]

- *Modified to form functional protein, e.g. proteolysis/cleavage of polypeptides OR biochemical modification of proteins, e.g. glycosylation*
- *Phosphorylation of proteins to increase or decrease its function*
- *Transportation of proteins to target destinations in the cell where it functions*
- *Protein degradation allows control of protein activities*
- *Ubiquitination marks protein for degradation → proteasome*

- (b) (i) In a laboratory, it was found that a particular cell line has a mutation in the *ras* gene promoter, leading to high expression of *ras* protein. However, cells from this cell line do not develop into tumour. Explain.

[2]

- *Multi-step process for tumour formation*
- *Requires tumour suppressor genes, e.g. p53 tumour suppressor to be mutated / loss of function mutation*
- *Genes whose protein products inhibit cell division during DNA damage and help prevent inappropriate cell cycle progression*
- *Result in arrest of cell division to allow more time for the cell to DNA repair and undergo apoptosis*

- (ii) Other than the reason stated in (i), explain other causes that can lead to high level of expression of *ras* protein.

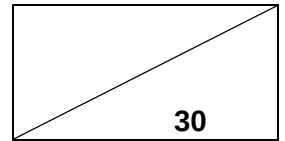
[2]

- *Mutation within exon of proto-oncogene*
 - o *Changes in codon / amino acid sequence in protein*
 - o *3D structure of protein changed and protein becomes hyperactive / resistant to degradation*
- *Chromosomal translocation of proto-oncogene to another site*
 - o *Under control of active promoter / control element*
 - o *Ref: gene is overexpressed, forming excessive amount of protein*
- *Gene amplification of proto-oncogene*
 - o *Error in cytokinesis / unequal crossing over / DNA slippage generates multiple copies of the gene*
 - o *Ref: to gene is overexpressed, forming excessive amount of protein*
- *Viral integration of oncogenes into genome*
 - o *Ref: expression of protein which is hyperactive / resistant to degradation*

[Total: 10 m]

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Section B

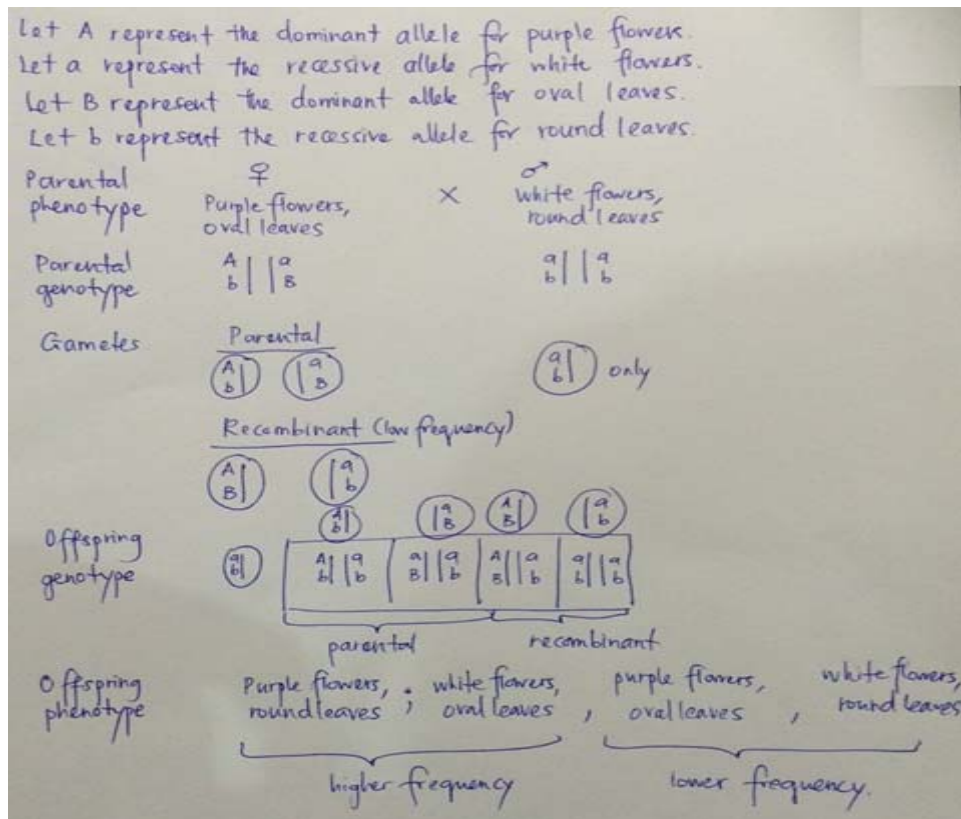
Answer **all** the questions in this section.

- 4 A botanist during his investigation on the exotic *Mimosa pudica*, crossed a pure-bred plant with purple flowers and round leaves with a pure-bred plant with white flowers and oval leaves. All the F₁ generation had purple flowers and oval leaves.

A test cross was then made between the F₁ plant and a pure-bred plant with white flowers and round leaves. The phenotypes of 250 offspring of the test cross were recorded and shown below.

Purple flowers, oval leaves : 21
 Purple flowers, round leaves : 102
 White flowers, oval leaves : 110
 White flowers, round leaves : 17

- (a) Using A/a symbols to represent locus for flower colour and B/b symbols to represent locus for leaf shape, draw a genetic diagram to explain the results of the test cross. [4]



- A Both parental phenotypes and genotypes are correct. [1]
- All possible gametes from each parent are correct. [1]
- Genetic diagram correctly shows the possible combinations of gametes. [1]
- The offspring phenotypes and genotypes are correct with reference to higher frequency for parental and lower frequency for recombinant combination. [1]

(b) Explain why the expected dihybrid test cross ratio of 1:1:1:1 was not observed. [2]

- The 2 genes are **linked** and do **not segregate independently** at anaphase of meiosis;
- **recombinant phenotypes** may arise due to **crossing over** during prophase 1 of meiosis by chance, though there is a **lower frequency of recombinants** due to **lower chances of occurrence for crossing over**, proportions lower for recombinants than parental phenotypes;

(c) In another investigation on the height of the *Mimosa pudica*, the botanist crossed two plants with different stem heights. The results of the cross are shown in Figure 4.1. [1]

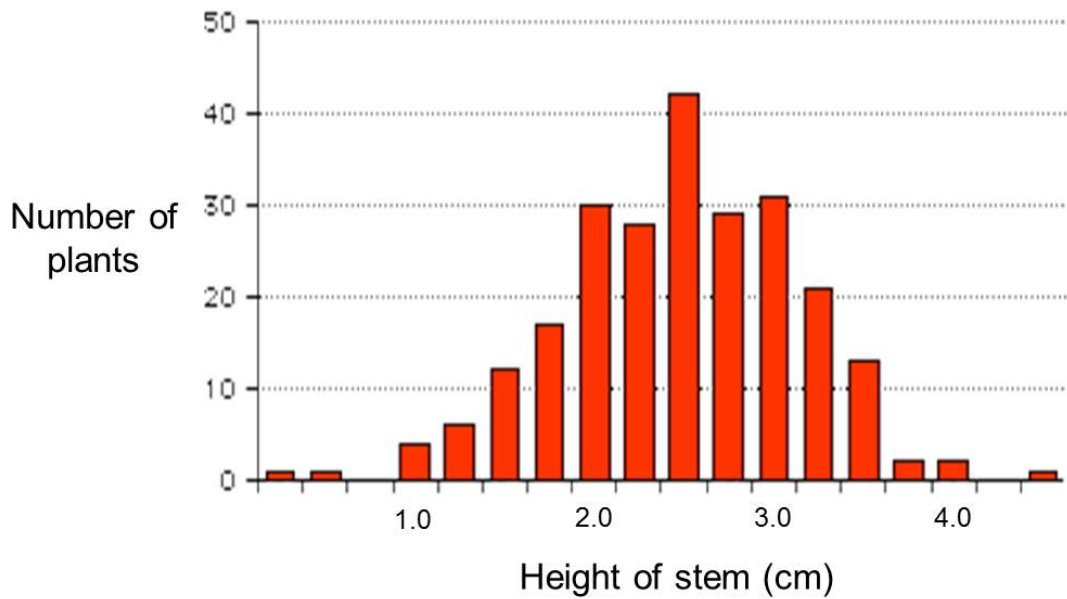


Figure 4.1

(i) State the type of variation shown in Fig. 4.1?

- *Continuous variation*

(iv) Explain why there is a range of phenotypes observed for stem height in *Mimosa pudica*.

[2]

- *Due to additive effect of polygenes, where the stem height is the combined effects of many genes, with each gene having a little overall effect on the stem height;*
- *The stem height might also be affected by environment, i.e. climate, nutrition available;*

[Total: 10 m]

- 5 The Middle East Respiratory Syndrome (MERS) is a fatal viral respiratory illness caused by a strain of coronavirus called MERS-CoV. The MERS-CoV is a single-stranded RNA virus. Through its spike proteins, this virus infects cells in the respiratory tract in humans, leading to symptoms of fever, cough and shortness of breath, and eventually death.

Figure 5.1 shows a model of the MERS-CoV structure.

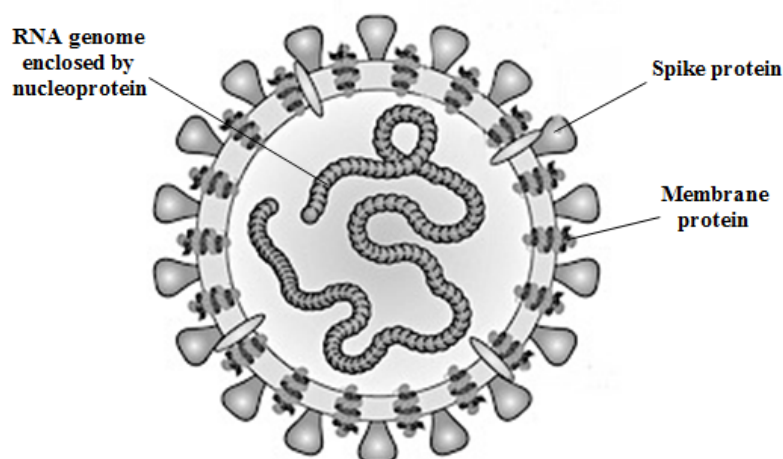


Fig. 5.1

- (a) Explain why viruses such as the coronavirus mutate faster than DNA viruses.

[2]

- *RNA viruses have higher rates of mutation because RNA-dependent RNA polymerase that synthesized RNA genome does not have the proof-reading capability that DNA polymerase synthesizing DNA genome of DNA viruses have;*
- *[Ref to appropriate enzymes – 1m, Ref to proof-reading – 1m]*

Researchers have recently discovered that the entry of MERS-CoV into its host cells is similar to how Human Acquired Immunodeficiency Virus (HIV) enters its host cells.

Suggest how MERS-CoV genome enters its host cells.

[2]

- *Spike protein/ surface protein on viral envelope bind to specific receptors on host cell membrane;*
- *promoting fusion of MERS-CoV viral envelope with host cell membrane surface releasing the RNA genome into the host cell cytoplasm;*
- *Reject: Reference to gp 120/41*

It was previously discovered by the researchers that the cleavage site on the spike protein of MERS-CoV is activated by a host cell protein that subsequently mediates

viral entry into host cell. Furin was recently found to be the protein that activates cleavage sites on MERS-CoV.

In a separate experiment, the researchers investigated the cellular furin content and the infection efficiency of MERS-CoV on different cell lines.

Figure 5.2 shows the relative furin abundance in different cell lines in the host organism. Figure 5.3 shows the Mers-CoV infection efficiency in the various cell lines investigated.

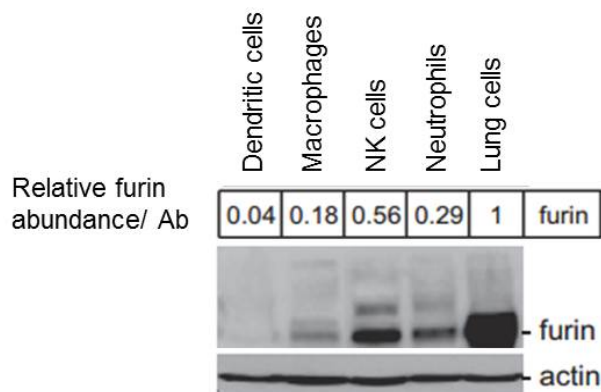


Fig. 5.2

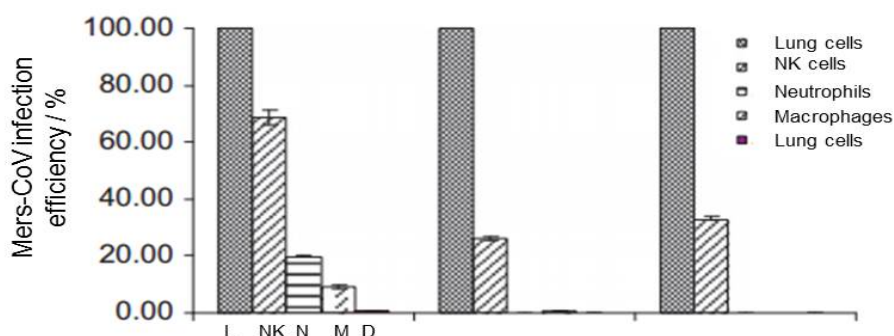


Fig. 5.3

- (c) With reference to Figures 5.2 and 5.3, explain what can be concluded regarding the cellular furin content and the infection efficiency of MERS-CoV on different cell lines.

[3]

- The higher the cellular furin content, where dendritic cells have the lowest furin content at 0.04 Ab to the lung cells that have highest furin content at 1Ab, the higher the Mers-CoV infection efficiency where that in dendritic cells is 0% and that in lung cells is 100%;; [Trend – 1m, Data quoted -1m]
- suggesting that host cell lines/ types with increased furin content facilitate the activation of viral spike protein increasing the infection efficiency of Mers-CoV;

Researchers predict that MERS-CoV is infectious because as it mutates, it is able to bind to different host cell proteins for activation of cleavage site, leading to viral entry.

(d) Explain the phenomenon that allows MERS-CoV to bind to different proteins for viral entry.

[2]

- *Antigenic drift, where mutations causing changes in the cleavage sites on spike protein, allow the virus to bind to different proteins on host cell surface for viral entry;*

(e) Explain why MERS-CoV is considered to be not only a parasite but an obligatory one.

[2]

(Obligate)

- *Requires host cell to reproduce OR*
- *Lacks many of the enzymes and /or organelles necessary for protein synthesis/ ATP generation / metabolism / DNA replication and requires such resources from host;*

(Parasites)

- *Takes over/ takes control/hijacks host cell metabolic machinery for the production of progeny viruses*

[Total: 10 m]

- 6 Fig. 6.1 shows the sequence of events in the insulin signaling pathway in a muscle cell.

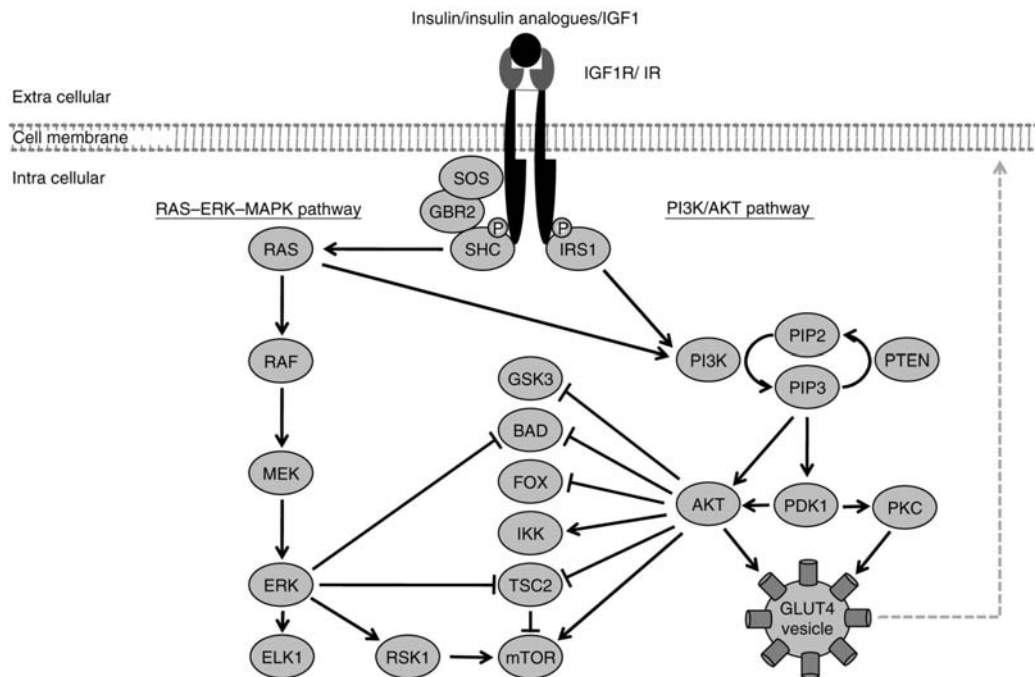


Fig. 6.1

- (a) With reference to Fig. 6.1, explain how a single binding event can trigger two different pathways? [2]

- *The dimerised tyrosine kinase receptors can cross-phosphorylate each other on multiple tyrosine residues;*
- *Each of these tyrosine residues can bind specific relay proteins, in this case SHC and IRS-1 and activate them, subsequently triggering two different downstream pathways;*

- (b) With reference to Figure 6.1, explain how activation of IRS-1 can lead to an increase in glucose permeability of muscle cells? [3]

- *IRS-1 activates PI3K which in turn converts PIP2 to PIP3;*
- *PIP3 activates AKT/ PDK1 which in turn activates PKC;*
- *which result in vesicles containing / carrying GLUT 4 glucose transporters (Reject: glucose channels) to move / migrate (Reject: diffuse) towards plasma membrane and fuse (Reject: bind) with the plasma membrane (or accept references to exocytosis) to incorporate / insert into or embed within the plasma membrane, thus increasing glucose permeability of muscle cells;*

- (c) Describe one other mechanism by which IRS-1 can lead to regulation of blood glucose levels. [2]

- Activated IRS1 activates other intracellular signaling proteins that include activating glycogen synthase;
- that converts glucose into insoluble glycogen, reducing blood glucose levels;

When insulin levels in the environment remain high for several hours, cells usually undergo desensitization, such that they no longer respond to that concentration of hormone. This prevents excessive and prolonged receptor activity.

- (d) Suggest how these cells can decrease the stimulation of receptor molecules by the presence of high concentration of insulin. [2]
- *Endocytosis (receptor mediated), reducing the number of available cell surface membrane receptors;*
 - *Less ligands binding to receptors, less stimulation of cell signaling pathway;*
- (e) Describe two advantages of a cell signalling system. [2]
- *Signal amplification is possible - very low concentration of a signal molecule is sufficient to produce a much greater cellular response;*
 - *Cells have the ability to regulate or control the response. For e.g., termination of the signal by endocytosis of ligand-receptor complex to prevent further ligand-receptor interaction;*
 - *Cellular response is specific depending on the type of proteins present in the cell;*
 - *Binding of a molecule to receptors on the cell membrane is able to activate genes in the nucleus (within the cell);*
 - *Many cells can be activated simultaneously by binding of ligand molecules to their receptors;*
 - *A single signal molecule is able to trigger numerous cellular reactions at once;*

[Any 2]

[Total: 11 m]

Name: _____ Class: 2bi2___ / 2IPbi2___

20

Section C

Answer **all** the questions in this section.

- 7 (a) Acetylcholine is a neurotransmitter that binds to postsynaptic membrane and stimulates the production of nerve impulses. GABA is another neurotransmitter that is produced by certain neurones in the brain and spinal cord. It binds to postsynaptic membrane and inhibits the production of nerve impulses.

Fig. 7.1 shows a synapse involving three neurones

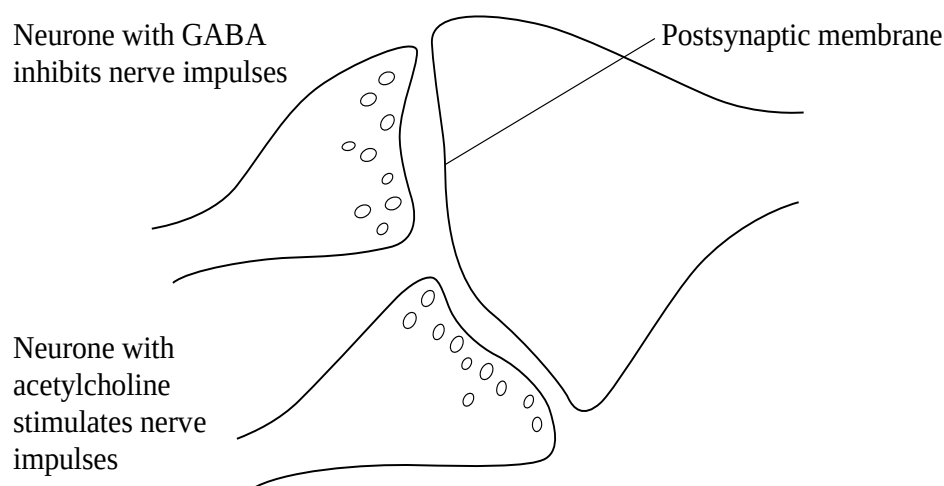


Fig. 7.1

- (i) Describe the sequence of events leading to the release of acetylcholine.

[3]

- *Action potential or depolarization occurs at the presynaptic terminal membrane and causes voltage-gated Ca^{2+} channels to open*
- *Influx of Ca^{2+} ions into the synaptic knob causes the synaptic vesicles to move towards and fuse with the presynaptic membrane*
- *Discharge of acetylcholine into the synaptic cleft via exocytosis*

- (ii) State and describe the function of one type of protein molecule on the post-synaptic membrane.

[3]

- *Ligand-gated ion channel*
- *Binding of ligand/neurotransmitter*
- *Causes conformational change that open the channel portion of the protein*
- *Allowing ions to cross the membrane*

OR

- *Acetylcholinesterase*
- *Breaks down acetylcholine*
- *Into choline and acetate which can then be reused*

(v) The binding of GABA to receptors on post-synaptic membrane causes negatively charged chloride ions to enter post-synaptic neurone. Explain how this will inhibit production of nerve impulses by post-synaptic neurone.

[2]

- *Inside of the neurone, will become more negatively charged (hyperpolarized) and does not reach threshold potential*
- *Action potential is not produced and depolarisation does not occur / reduces the effect of sodium ions entering*

(vi) Epilepsy may result when there is increased neuronal activity in the brain. One form of epilepsy has been linked to an abnormality in GABA receptors. Suggest how an abnormality in GABA receptors may result in epilepsy.

[2]

- *Abnormality in GABA receptor will result in receptors that have different tertiary / 3D structure*
- *Shape is not complementary*
- *GABA cannot bind to GABA receptors*
- *Inhibition of neuronal activity does not occur (and chloride ions do not enter)*

[Total: 10 m]

- 8 Mitochondrial DNA analysis of various rodents was done and their phylogenetic relationships represented by Fig. 8.1

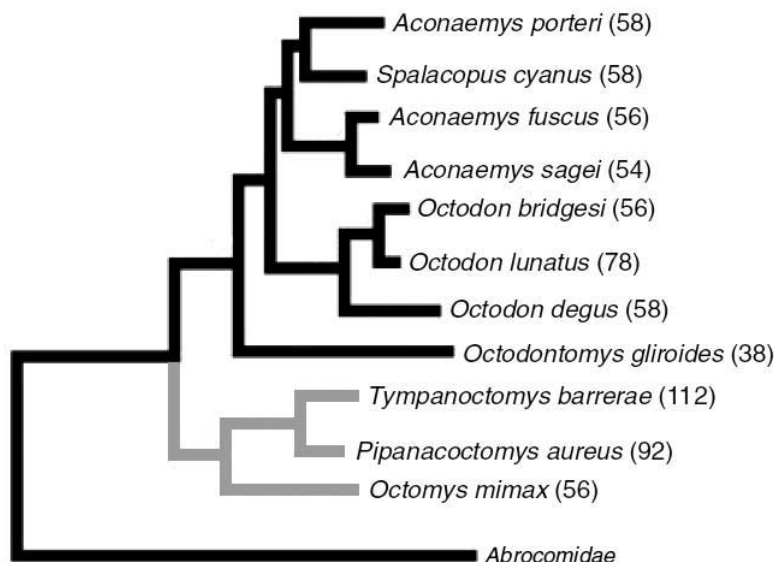


Fig. 8.1

- (a) (i) With reference to Fig. 8.1, state which species is most closely related to *Octodontomys gliroides*.

[1]

- *Octodon degus*

- (a) Suggest why mitochondrial DNA was used to study the phylogenetic relationships between the various species of rodents.

[2]

- Mitochondrial DNA is a homologous molecule/nucleotide sequence that is found in the mitochondria of every eukaryotic organism.
- Codes for proteins that are involved in important function and genetic differences are less common and hence easier to analyse for true evolutionary relationships.

- (b) Explain how DNA analysis of mitochondrial DNA can provide evidence for the relationships shown in Figure 8.1.

[1]

- DNA-DNA hybridization (ottwe) can be used to examine the phylogenetic relationships between species.
- The more complementary the nucleotide sequences, the more closely related they are. [A: reverse argument]

OR

- DNA analysis / compare sequence of mitochondrial DNA of all species of rodents.
- The greater the differences in nucleotide sequence, the more distantly related the species are. [A: reverse argument]

(c) Explain the difference between classification and phylogeny.

[2]

- *Classification refers to grouping organisms based on similar characteristics*
- *Characteristics may be analogous and not homologous*
- *Phylogeny involves grouping organisms based on evolutionary relationship*
- *Similarity is due to inheritance from common ancestry*

(d) Explain two advantages of using molecular evidence compared to anatomical homology in establishing phylogeny.

[2]

- *Objective*
- *Quantifiable*
- *Use of phenotypically non-visible characteristics*
- *Compute small changes*

(e) Cases of mammalian tetraploidy are rare. The Plains Viscacha Rat, *Tympanoctomys barrerae*, is one of the few polyploid mammals in existence and it is tetraploid (4n) with 112 chromosomes. It is only found in the desert scrublands of the central western plains in Argentina. It was hypothesised that *Tympanoctomys barrerae* evolved from the same common ancestor as *Octomys mimax*, a diploid rodent with 56 chromosomes (2n). *O. mimax* inhabits rocky desert and semi-desert environments up to 800 metres above sea level.

Explain how tetraploidy may have first developed in the ancestral rat population.

[2]

- *Spindle fibres may not have formed during prophase I of meiosis*
- *Cytokinesis may not have occurred during meiosis, and therefore the number of chromosomes remain in the diploidy state*
- *The fusion of two diploid gametes could have given rise to the tetraploid rat*

[Total: 10 m]

Section D

Answer **all** parts of this question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Explain how the structure of phospholipids facilitates the process of chemiosmosis in the chloroplast.. [10]

- *Phospholipid = glycerol + 2 fatty acids + phosphate group (joined by ester linkages);*
- *Amphipathic → phosphate head is hydrophilic and hydrocarbon tail is hydrophobic;*
- *Phospholipid arranges themselves in aqueous envt / to form bilayer;*
- *Enable proteins to be embedded;*
- *Held in place via hydrophobic interactions between fatty acid tail and proteins;*
- *Named examples: ETC and ATP synthase;*
- *Named thylakoid membrane in chloroplast;*
- *Enclosing thylakoid space; (award only when students describe the structure)*

[Max 5]

- *Hydrophobic hydrocarbon core/tail impermeable to polar ions/protons;*
- *Allows building of proton pool/proton reservoir in thylakoid space / generation of proton gradient across thylakoid membrane; R! thylakoid gradient in thylakoid space;*
- *Protons diffuse into stroma through ATP synthases to produce ATP;*
- *Phospholipid contains unsaturated fatty acid tail;*
- *Kink prevents tight packing of phospholipid;*
- *Maintain fluidity of thylakoid membrane;*
- *Allows proper function of transmembrane proteins;*
- *Such as movement of electron carrier proteins;*

[Max 5]

@ 1m, max 10

(b) Compare and contrast aerobic and anaerobic respiration.

[10]

Similarities: (max 5)

Purpose: Both serve to generate energy in terms of ATP molecules to sustain energy-requiring activities.

Common process: Glycolysis is common to both, whereby glucose is broken down into pyruvate, giving rise to a net gain of 2 ATP molecules, and producing 2 NADH molecules as well in the cytoplasm.

Involvement of hydrogen acceptor: Both needs NAD⁺ as a hydrogen acceptor / co-enzyme for dehydrogenase to complete dehydrogenation reactions in glycolysis (from glyceraldehydes 3 phosphate to 1,3-bisphosphoglycerate)

Both have a process to regenerate the hydrogen acceptor NAD⁺ to support further glycolysis (Note: student can also put this as a difference, if they describe the mechanism involved in regenerating NAD⁺)

OR

Both only harvest a portion of the chemical energy stored in glucose molecule, with a large amount of energy lost as heat.

OR

any other valid points

(Note: description of common process glycolysis can only be awarded once, any other detailed descriptions of steps in glycolysis won't be awarded any marks)

Differences (max 5): (use a table to present your answer)

Feature	Aerobic respiration	Anaerobic respiration
Location	In cytoplasm and mitochondria	Only in cytoplasm
Hydrogen acceptor	NAD ⁺ and FAD	Only NAD ⁺
Stages involved in generating ATP	Glycolysis, link reaction, Krebs cycle and oxidative phosphorylation	Only glycolysis and fermentation
Mechanism of ATP synthesis	Direct ATP synthesis via Substrate level phosphorylation during glycolysis and krebs cycle; Oxidative phosphorylation through electron transport chain and chemiosmosis.	Only substrate-level phosphorylation during glycolysis
Number of ATP synthesized per glucose molecule	38 (4 via substrate-level phosphorylation and 34 via oxidative phosphorylation)	Only 2 via substrate-level phosphorylation
Final products	CO ₂ , ATP, H ₂ O	Lactate, ATP OR CO ₂ , ethanol and ATP
Type of chemical reactions involved in breaking down	Dehydrogenation, Oxidative decarboxylation,	Only dehydrogenation

<i>glucose</i>		
<i>Mechanism where hydrogen acceptor is regenerated</i>	<i>Via electron transport chain during oxidative phosphorylation, whereby NADH and FADH₂ donate the electrons and protons to the electron carriers</i>	<i>Via reduction reaction (ethanol to ethanol; pyruvate to lactate) during fermentation</i>
<i>Involvement of O₂</i>	<i>Yes, as the final electron acceptor in the ETC (Reject: just 'Yes' without elaboration)</i>	<i>No.</i>

[Total: 10 m]

10 (a) Define the terms gene and allele and explain how they differ. [4]

- *Gene is a heritable factor / unit of inheritance*
- *Gene is composed of DNA*
- *Gene controls a specific characteristic / codes for a polypeptide / protein*
[Max. 2 marks]
- *Allele is a form of a gene*
- *Alleles of a gene occupy the same gene locus / same position on chromosome*
- *Occur in pairs in a diploid cell*
- *Causes slightly different phenotypes for the same characteristic*
- *Alleles differ (from each other) by one or a small number of bases / base pair(s)*
[Max. 2 marks]

(b) Describe the chemical structure of the gene. [8]

- *Genes are composed of units of DNA*
- *Name 3 out of the 4 nitrogenous bases (A, T, C, G)*
- *N bases are paired (A-T, G-C)*
- *Components of nucleotide / DNA*
- *A / G bases are purines and T / C are pyrimidines*
- *DNA is a double stranded helix*
- *H-bonds between N bases in base-pairing*
- *The two DNA strands are antiparallel*
- *DNA codes for amino acid sequence in proteins*
- *Each gene has a start and stop nucleotide sequence*

(c) Distinguish between mitosis and meiosis [8]

	<i>Mitosis</i>	<i>Meiosis</i>
1	<i>Dividing cells can be diploid or haploid</i>	<i>Dividing cells are diploid</i>
2	<i>Occurs in somatic cells in all parts of the body.</i>	<i>Occurs only in certain cells present in the gonads.</i>
3	<i>DNA replicates once during the S phase of interphase, and nucleus divides once.</i>	<i>DNA replicates once during the S phase of interphase but there are two successive nuclear divisions.</i>
4	<i>Homologous chromosomes do not associate during prophase.</i>	<i>Homologous chromosomes associate to form bivalents during prophase I</i>
5	<i>Chiasmata are never formed, and crossing over never occurs.</i>	<i>Chiasmata may form, and crossing over may occur during prophase I</i>
6	<i>Chromosomes form a single row at the equator of the spindle during metaphase.</i>	<i>Chromosomes form two rows at the equator of the spindle during metaphase I.</i>
7	<i>Chromatids move to opposite poles of spindle during anaphase.</i>	<i>Chromosomes move to opposite poles of spindle during anaphase I.</i>

8	<i>Daughter cells have the same number of chromosomes as the parent cell.</i>	<i>Daughter cells have only half the number of chromosomes found in the parent cell.</i>
9	<i>In the absence of mutation, daughter cells are genetically identical to parental cell.</i>	<i>Daughter cells are genetically different from parental cell.</i>
10	<i>Two daughter cells are formed.</i>	<i>Four daughter cells are formed.</i>

[Total: 10 m]